Biomedical Signal Processing and Control 129 (2027) 111507

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

DERMIS: An explainable dual-branch ensemble framework for robust melanoma detection

Ashwin Singh Slathia, Saksham Anand, Vasu Arya, Abhiram Sharma, Ayush Rathi, Pattabiraman V., R. Parvathi *

*School of Computer Science and Engineering, * Centre for Advanced Data Science, Vellore Institute of Technology Chennai, Chennai, 600127, Tamil Nadu, India*

ARTICLE INFO

Keywords:
Explainable AI
Dual-branch fusion
GAN augmentation
Biomedical image analysis
Medical signal processing
Deep learning
DERMIS framework

ABSTRACT

Accurate melanoma lesion classification is inherently linked with early diagnosis and improved prognosis. However, class imbalance, non-interpretable deep models, and poor feature extraction abilities restrict their clinical use. To address these limitations, *Dual-branch Ensemble for Robust Melanoma Identification System* or *DERMIS* is proposed, which integrates DenseNet201 and MobileNetV3-Large with an attention-based fusion approach to surpass state-of-the-art models in all evaluation measures. Synthetic malignant images are crafted by Generative Adversarial Networks (GAN)-based image augmentation with balanced class representation and improved model generalizability. Aggressive data preprocessing operators like normalization, random rotation, flipping, zoom, and shearing improve data quality and model robustness. Explainable AI tools, i.e., LIME, SHAP, and Grad-CAM, facilitate interpretable model decision-making with comprehensive interpretability analysis. DERMIS superiority over benchmark models is proven through exhaustive comparative analysis with 95.11% accuracy, 96.66% precision, 93.13% recall, and 94.86% F1-score, outperforming EfficientNetB3, BeIt, InceptionResNetV2, DenseNet201, ResNet101, CoaTNet, and MobileNetV3 architectures significantly. The model was trained and validated against dataset containing 10,605 RGB images with binary melanoma classification using an 80:20 training–testing split. Exhaustive ablation studies confirm the synergistic benefits of the dual-branch fusion strategy over single baseline models. Grad-CAM heatmaps have the best clinical interpretability among LIME and SHAP, accurately localizing pathologically important features like asymmetric morphology, irregular borders, and abnormal pigmentation patterns. DERMIS is an important breakthrough in melanoma detection with automated methods, offering best trade-off between diagnostic accuracy, computational efficiency, and clinical interpretability.

1. Introduction

Melanoma is one of the most dangerous forms of skin cancer, with incidence rates rising at alarming levels globally, particularly among youth. The American Cancer Society reports approximately 99,780 new melanoma cases annually, with 7650 fatalities. Early detection is crucial localized melanoma has a five-year survival rate exceeding 99%, whereas advanced metastatic cases drop to just 27%. This disparity highlights the critical need for accurate and timely diagnostics in clinical dermatology.

Conventional diagnosis relies on dermatological examination, which occasionally aided by dermoscopy for enhanced visualization. However, this approach is limited by inter observer variability, subjective judgment, and limited access in low-resource settings. Diagnostic accuracy among dermatologists ranges from 65% to 85%, depending on expertise and complexity. Rising patient volumes and a global shortage of specialists emphasize the need for robust computer aided diagnostic (CAD) systems to support clinical workflows.

Recent advances in deep learning have transformed medical image analysis, which offers improved diagnostic accuracy and consistency. Convolutional Neural Networks (CNNs) have shown remarkable success in radiology, pathology, and dermatology applications. Their adoption in skin lesion classification has yielded promising results, occasionally surpassing expert-level performance. However, certain issues remain, such as the complexity of architecture, class imbalance, and the absence of clinical interpretability.

Melanoma detection via dermoscopic images presents distinct hurdles: lesion heterogeneity, imaging variability, and imbalanced datasets where benign cases vastly outnumber malignant ones can impair model performance. Furthermore, the "black-box" nature of deep learning models raises concerns over clinical trust and regulatory acceptance.

* Corresponding author.
E-mail address: parvathi.r@vit.ac.in (R. Parvathi).

https://doi.org/10.1016/j.bspc.2026.111507
Received 29 October 2025; Received in revised form 6 April 2026; Accepted 14 September 2026
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Various CNN architectures, including ResNet, DenseNet, EfficientNet, and MobileNet, have been assessed for skin lesion classification. While each excels in certain aspects, their limitations can be complementary. DenseNet captures fine-grained textures, while MobileNet offers computational efficiency for mobile use. Fusion or ensemble techniques can unify these advantages for better results.

Explainable AI (XAI) is gradually being incorporated into medical diagnostics to meet the pressing need for model transparency. To this end, the medical community needs interpretable results to confirm the outcome, which will enhance trust and ensure that the model's logic is consistent with medical knowledge. LIME, SHAP, and Grad-CAM offer complementary views of interpretability, each with its own advantages in the medical domain. SHAP has also been successfully used in large-scale computational systems to enhance decision transparency, showing its applicability in other domains outside medical image analysis scenarios [1].

Although there has been significant advancement in deep learning-based skin lesion analysis, there are still three major gaps: the current over-reliance on individual model architectures, the lack of utilization of sophisticated augmentation techniques such as GANs for class balancing, and the lack of comparative analysis of XAI techniques in dermatology.

To address these gaps, this paper proposes DERMIS (Dual-branch Ensemble for Robust Melanoma Identification System), a novel CNN fusion framework integrating DenseNet201 and MobileNetV3-Large with an attention based fusion module. GAN based synthetic augmentation handles class imbalance, while a comprehensive XAI framework (LIME, SHAP, Grad-CAM) ensures interpretability and clinical transparency. Unlike simple ensembles, DERMIS employs a learnable attention-based dual-branch fusion that adaptively integrates complementary DenseNet201 and MobileNetV3-Large feature representations, together with a GAN-based class-balancing strategy applied exclusively to the training split. Furthermore, interpretability is systematically evaluated using multiple XAI techniques (Grad-CAM, LIME, and SHAP), enabling reproducible improvements in diagnostic performance and clinically meaningful model transparency beyond conventional model averaging.

The major contributions of this research include:

1. A dual-branch fusion model leveraging complementary CNN strengths.
2. GAN-based augmentation to address class imbalance.
3. Integration of XAI techniques to enhance clinical trust and model transparency.
4. Demonstrated superior performance compared to state-of-the-art baselines.

The proposed *DERMIS* model achieves outstanding performance: **95.11%** accuracy, **96.66%** precision, **93.13%** recall, and **94.86%** F1-score, setting a new benchmark in automated melanoma detection from dermoscopic images.

2. Literature review

Skin cancer detection has been widely studied with deep learning and AI methods improving classification accuracy. CNN models have been used for lesion classification. An interpretable CNN model addressed class imbalance and explainability achieving 82% accuracy as studied by K Mridha et al. [2]. Another study used Swin Transformer and Dense Group Shuffle Non-Local Attention Network to enhance feature representation reaching 94.21% accuracy as explored by R Karthik et al. [3]. Generative AI techniques such as GANs and Enhanced Super-Resolution GAN (ESRGAN) improved classification accuracy to 96% with a hybrid VGG19-Support Vector Machine (SVM) model as presented by M Saeed et al. [4]. A metadata-based deep learning approach combined images and patient demographics enhancing classification accuracy as studied by A G C Pacheco et al. [5]. A handheld microwave system detected in vivo skin lesions by analyzing dielectric properties distinguishing benign from malignant lesions as shown by S A R Naqvi et al. [6]. Some studies combined segmentation and classification. A mutual bootstrapping model segmented and classified skin lesions simultaneously achieving high accuracy with convolutional networks as studied by Y Xie et al. [7]. Another approach used transfer learning with InceptionV3 and Xception to develop an ensemble model improving classification accuracy to 98% as shown by S Chatterjee et al. [8]. RNA-Seq and microarray data integration improved diagnostic insights identifying gene markers for different skin cancer types as presented by J M Gálvez et al. [9]. A CNN-based framework combined segmentation and classification reaching 98% accuracy improving diagnostic efficiency as studied by A A Adegun et al. [10]. Teledermatology-based classification models utilized DenseNet201 with transfer learning for multi-class lesion detection as reported by M A Khan et al. [11]. DenseNet201 has also been integrated within hybrid diagnostic frameworks, such as its combination with InceptionV3 for cervical cancer classification, further validating its effectiveness for biomedical feature extraction [12].

Feature extraction played a key role in enhancing classification accuracy. A study proposed an automated non-invasive real-time system for melanoma detection using shape color and texture features achieving 97.5% accuracy as shown by O Abuzaghleh et al. [13]. Another approach used Gaussian filtering K-means clustering and SVM for melanoma and nevus classification achieving 96% accuracy as explored by M Q Khan et al. [14]. DeepSkin a CNN model trained on the HAM10000 dataset employed ResNet50 and DenseNet169 improving classification results as reported by H L Gururaj et al. [15]. A four-class classification strategy distinguished melanocytic and nonmelanocytic lesions achieving 90.48% detection accuracy for melanomas as presented by K Shimizu et al. [16]. Region-of-interest-based transfer learning improved melanoma detection by isolating lesion features enhancing accuracy as shown by R Ashraf et al. [17].

Hybrid deep learning models combined CNNs with advanced architectures. A deep ensemble model DCENSnet improved classification accuracy through dropout-based feature learning achieving 99.53% accuracy as studied by D Chanda et al. [18]. A GAN-based augmentation model enhanced multi-class classification mitigating class imbalance and improving specificity to 98.34% as reported by Q Su et al. [19]. A 3D reconstruction-based system extracted tumor depth features improving melanoma detection accuracy to 98% as shown by T Y Satheesha et al. [20]. Unit-Vise a deep residual learning model reduced overparameterization issues improving classification accuracy to 98.05% as presented by I Razzak et al. [21]. FixCaps an improved capsule network applied large-kernel convolutions and attention mechanisms increasing detection accuracy to 96.49% as explored by Z Lan et al. [22]. Similar dermatological classification frameworks have combined GAN augmentation with SHAP and LIME-based interpretability to enhance both performance and model transparency [23].

Segmentation and classification models continued evolving. A hybrid CNN model with an SVM classifier improved dermoscopic image classification achieving higher accuracy than traditional CNNs as studied by D Keerthana et al. [24]. A Stokes-Mueller decomposition-based model extracted optical parameters from skin lesions improving classification accuracy as explored by T N Luu et al. [25]. A web-based adaptive system used machine learning for risk assessment optimizing classification with Naïve Bayes and k-NN models as presented by T Y Yang et al. [26]. Genetic programming-based classifiers extracted local and global features providing high interpretability for dermatologists as studied by Q Ul Ain et al. [27]. A multi-stage transfer learning approach combined five different CNN architectures improving classification accuracy through ensemble learning as reported by A T Ibrahim et al. [28].

Recent works combined AI-based optimization methods. A full-resolution CNN model with dynamic graph cut algorithm enhanced segmentation with 97.986% accuracy as researched by D Adla et al. [29].

Table 1
Comprehensive comparison of representative skin cancer classification approaches.

Study/Model	Model type	Ensemble	GAN	XAI	Accuracy (%)	Key limitation
Mridha et al. [2]	CNN	No	No	Yes	82.0	Low accuracy; limited generalization.
Karthik et al. [3]	Transformer + Attention	No	No	No	94.21	Computationally heavy; no explainability integration.
Saeed et al. [4]	CNN + SVM Hybrid	No	Yes	No	96.0	Focused on augmentation (GAN/ESRGAN); lacks systematic XAI.
Chatterjee et al. [8]	Transfer learning ensemble	Yes	No	No	98.0	High inference cost; no interpretability analysis.
Su et al. [19]	CNN + GAN augmentation	No	Yes	No	98.34	Imbalance mitigation only; lacks ensemble fusion and XAI.
Khater et al. [32]	CNN + XAI	No	No	Yes	94.0	Focus on explainability; no fusion or GAN balancing.
Yu et al. [33]	Weighted CNN ensemble	Yes	No	No	94.49	Ensemble improves accuracy but increases computational cost.
DERMIS (Proposed)	Dual-branch fusion CNN	Yes	Yes	Yes	95.11	Higher training complexity compared to lightweight models.

A hyperspectral imaging method employed SVM and XGBoost in classification showing potential in spectral analysis as demonstrated by B Petracchi et al. [30]. Deep learning models based on augmented intelligence enhanced feature extraction and improved classification in industrial IoT applications as stated by A Kumar et al. [31]. An explainable AI model was used with SHAP and permutation importance to define the key features in melanoma classification with 94% accuracy as reported by T Khater et al. [32]. An ensemble of CNNs using test-time augmentation and a weighted combination enhanced the accuracy of classification to 94.49% indicating the usefulness of ensemble strategies as investigated by K Yu et al. [33].

Classification was further enhanced by multi-modal imaging methods. A multimodal imaging screening system incorporating thermal and visible light imaging classified lesions with 98% sensitivity as investigated by S Diaz et al. [34]. The use of a CNN model that combines dermoscopic images and metadata increased the classification accuracy by taking into account demographic factors as demonstrated by R Maron et al. [35]. S Mavaddati et al. [36] reported state-of-the-art classification results on the ISIC dataset with a deep learning model based on EfficientNet and DenseNet. A computerized classification system used feature fusion methods that enhanced melanoma detection accuracy to 99.6% as examined by U Sara et al. [37]. A double AMIS-ensemble deep learning model that is AI-driven outperformed traditional CNN-based methods in terms of speed and accuracy as introduced by K Sethanan et al. [38].

Several approaches were designed owing to the advances in deep learning. A CNN-based approach using residual learning further improved the classification process of skin cancer types by optimizing feature extraction approaches as proposed by R Dandu et al. [39]. A mobile-optimized lightweight model further offered a practical solution for real-time detection as proposed by T Khater et al. [40]. A hybrid deep learning approach combined Long Short-Term Memory (LSTM) and ResNet50 for classifying skin lesions with higher accuracy by minimizing feature redundancy as shown by S Mavaddati et al. [41]. AI-driven lesion segmentation models further improved classification efficiency refining image preprocessing techniques as reported by A Kumar et al. [42]. A genetic programming method applied adaptive region detection outperforming traditional feature extraction techniques as presented by Yu et al. [33].

Although previous researchers have shown remarkable performance with advanced CNNs, ensembles, and data augmentation techniques, these methods tend to tackle one of the challenges individually. DenseNet201-based feature extraction has also been combined with YOLOv7 detection mechanisms for automated diabetic ulcer and wound classification, highlighting its adaptability across clinical imaging applications [43]. Some of them focus on architectural depth or ensemble learning to improve accuracy, and others on mitigating class imbalance or explainability alone. Nevertheless, little attention has been given to coherent frameworks that are able to simultaneously capitalize on complementary feature fusion, strong imbalance processing, and systematic explainability at clinical relevance. In particular, comparative evaluation of explainable AI techniques and their localization reliability for dermatological decision support remains underexplored.

Although significant progress has been made in the classification of skin cancer, most of the existing approaches are based on single or computationally intensive ensemble models, making them less efficient and less generalizable. Class imbalance and overfitting remain among the most challenging problems, even with the use of sophisticated augmentation approaches. Furthermore, most of the high-performing models lack robust and clinically reliable interpretability. These issues provide a rationale for the proposed methodology in the subsequent section.

Algorithm 1 DERMIS Framework

Require: Dataset D with N labeled images, Pretrained DenseNet201, MobileNetV3-Large
Ensure: Trained FusionModel with optimized weights
Preprocessing
Resize images to 224×224
Apply augmentations: Flip, Brightness-Contrast, Rotation, Hue-Saturation
Normalize with mean = std = (0.5, 0.5, 0.5)
Split D into D_{train} and D_{val}
Model Definition
Extract $F_{\text{dense}}, F_{\text{mobile}}$ from pretrained backbones
$F = [F_{\text{dense}}, F_{\text{mobile}}]$
$F' = F \cdot \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot F))$
$Y = \text{Classifier}(F')$
Training Phase
Initialize optimizer (Adam, $\eta = 10^{-4}$), CrossEntropy loss, CosineAnnealingLR
for each epoch $e = 1$ to E **do**
  for each mini-batch $(X, Y) \in D_{\text{train}}$ **do**
    $(X_{\text{mix}}, Y_a, Y_b, \lambda) \leftarrow \text{CutMix}(X, Y)$
    $\hat{Y} = f(X_{\text{mix}})$
    $L = \lambda \cdot \mathcal{L}(\hat{Y}, Y_a) + (1 - \lambda) \cdot \mathcal{L}(\hat{Y}, Y_b)$
    Backpropagate and update weights
  end for
  Evaluate on D_{val}
  if no improvement for P epochs **then**
    Early stopping
  end if
end for
Save best weights
Evaluation Phase
for each $(X, Y) \in D_{\text{val}}$ **do**
  Apply 3× Test-Time Augmentation (TTA)
  $\hat{Y} = \frac{1}{3} \sum_{i=1}^{3} \text{Softmax}(f(X))$
end for
Compute final metrics
return Trained FusionModel

3. Methodology

This study introduces a generalizable binary skin lesion classification as benign or malignant. The novel DERMIS architecture, visually depicted in Fig. 2 and algorithmically described in Algorithm 1, forms

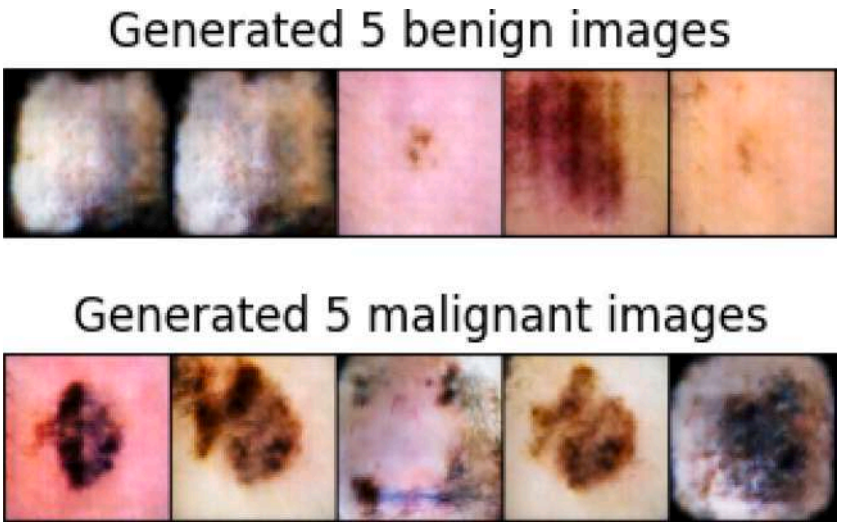


Fig. 1. GAN generated images.

the basis of this study. It includes the data pipeline, architectural framework, and innovations like GAN-based data balancing, dual-branch CNN fusion, and multi-perspective explainability integration.

3.1. Dataset acquisition and characteristics

The dermoscopic image data was downloaded from Melanoma Skin Cancer Dataset [44], which consists of 10,605 Red-Green-Blue (RGB) images with 5500 benign and 5105 malignant cases. This clinically typical distribution reflects dataset imbalance commonly faced in dermatology practice. To preserve diagnostic information such as vascular structure, lesion morphology, and pigmentation patterns all images were maintained in their original RGB color space.

3.2. Data preprocessing and augmentation

Generative Adversarial Networks (GAN) were used in an end to end preprocessing pipeline to create synthetic data. To address dataset imbalance, GAN-based augmentation is used to expand underrepresented classes. The GAN was trained separately on the training set only (avoiding test-set leakage), and up to $\tilde{2}5\%$ synthetic images were added relative to the original dataset. This ensured balanced representation without exceeding a safe augmentation ratio. A discriminator D(x) will separates real from artificial images, and a generator G(z) will projects random noise vectors z into image space. The objective function follows the minimax formulation:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}\left[\log\left(1 - D(G(z))\right)\right] \quad (1)$$

The generator used transposed convolutional layers with the batch normalization and ReLU activations, while the discriminator will use standard convolutional layers with leaky ReLU activations. Fig. 1 shows synthetically generated samples with realistic lesion morphology and color distribution.

The GAN training setup was well chosen to maintain stable adversarial learning and the synthesis of realistic melanoma images. The GAN was trained over 400 epochs, which provided sufficient adversarial interaction between the discriminator and the generator to learn fine-grained lesion structures without early convergence. The Adam optimizer was used with $\beta_1 = 0.5$ and $\beta_2 = 0.999$ to stabilize the updates and minimize oscillatory dynamics between the adversarial networks, which is a common choice in GAN training. The learning rates were asymmetrical, with the generator trained using a learning rate of 0.00005, whereas the discriminator was trained with a learning rate of 0.0002, allowing the discriminator to learn more robust gradients while avoiding generator instability and mode collapse. Diversity was balanced using a latent noise vector of 100 dimensions, and optimization complexity was controlled with a batch size of 64, which provided consistent gradient estimates and computational efficiency at 128×128 resolution. Training was performed exclusively on the training split to prevent test-set leakage.

To increase the diversity while preserving clinical realism, robust data augmentation techniques were methodically used. Pixel intensities were normalized to [0,1] by dividing by 255, clinical variability was simulated by the rotations (±30°), shifts (±0.2), and zooms (0.8–1.2). Horizontal and vertical flips will enlarges the dataset without adding artifacts, while shear transforms and brightness variation addressed perspective and illumination variations. These techniques reduced overfitting while improving the models resilience to actual imaging conditions.

Besides traditional augmentations and class balancing with GAN, CutMix is also used in training as illustrated in Algorithm 1. CutMix is a training method that creates mixed training examples, in which a patch of one image is replaced by that of another image and the labels of both images are then mixed together in a proportional way. The strategy is complementary to GAN augmentation, promoting more continuous decision boundaries between benign and malignant classes, as an extra regularization mechanism to curb overfitting, and more resilient to partial occlusions. Furthermore, CutMix makes explanations based on localization more stable and clinical-relevant by obstructing the model to focus on a single dominant feature.

3.3. Proposed model architecture

Model selection involved extensive testing of the most recent convolutional neural network architectures like EfficientNetB3, BeIt, InceptionResNetV2, DenseNet201, ResNet101, CoaTNet, and MobileNetV3-Large. The architectures were selected for certain strengths: EfficientNetB3 for computational efficiency, BeIt for vision transformer support, InceptionResNetV2 for multi-scale feature extraction, DenseNet201 for dense connections, ResNet101 for deep residual learning, CoaTNet for convolution and attention hybrid, and MobileNetV3-Large for light deployment. The proposed DERMIS architecture exploits complementary strengths by thoughtful combination of DenseNet201 and MobileNetV3-Large networks shown in Fig. 2. The dual-branch strategy takes advantage of high feature reuse ability of DenseNet201 using dense connections to allow thorough texture examination essential for melanoma detection, while MobileNetV3-Large offers computationally friendly feature extraction apt for clinical deployment environments.

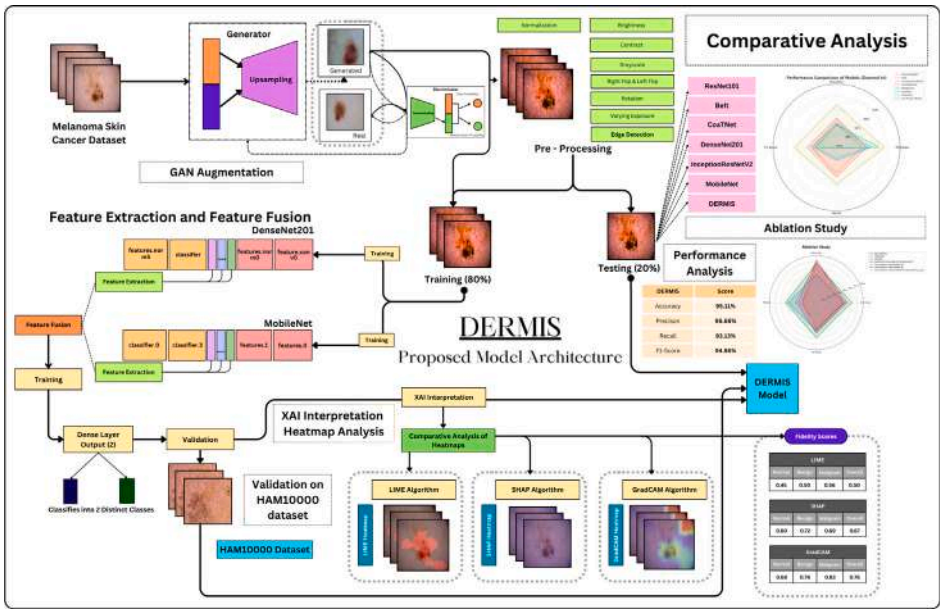


Fig. 2. DERMIS model architecture.

3.4. Fusion architecture implementation

The fusion mechanism operates through a sophisticated five-stage process of optimizing complementary feature utilization. Input images are first feature-extracted as a pre-trained DenseNet201 backbone, where hierarchical representations are increasingly accessed by convolutional layers. Global Average Pooling reduces spatial dimensions without losing feature intensity, which constructs a 1920-dimensional feature vector for each image.

The same input is passed through the MobileNetV3-Large pathway at the same time, yielding a 960-dimensional speed-optimized feature representation. The two complementary feature vectors are concatenated, providing a combined 2880-dimensional representation that retains both fine textural detail from DenseNet201 and spatially frugal patterns from MobileNetV3-Large. The attention-fusion mechanism takes advantage of a learnable feature contribution weighting policy to optimize feature contribution. The attention module estimates importance weights through:

$$\alpha = \sigma\left(W_2 \cdot \mathrm{ReLU}\left(W_1 \cdot f_{concat} + b_1\right) + b_2\right) \tag{2}$$

where f_{concat} represents the concatenated feature vector, W_1 and W_2 are learned weight matrices, b_1 and b_2 are bias terms, and σ denotes the sigmoid activation function. The refined feature vector is computed as:

$$f_{refined} = \alpha \odot f_{concat} \tag{3}$$

where $\odot$ represents element-wise multiplication, enabling selective emphasis of discriminative features while suppressing irrelevant information.

The head employs a multi-layer structure with progressive dimension reduction with Layer Normalization for training stability in the small batch situation and Dropout layers with rates 0.4 and 0.3 for regularization. The batch normalization and L2 regularization techniques were employed systematically to prevent overfitting but with the model remaining expressive.

3.5. Fine-tuning

Since the dataset size is more than 10,000 samples, all the network layers were kept unfrozen to enable end-to-end parameter optimization. The overall fine-tuning strategy enables learning pretrained ImageNet features to adapt to dermoscopic image features, optimizing low-level edge representation and high-level semantic information for melanoma classification.

3.6. Training configuration and optimization

The training process employed Label Smoothing Cross-Entropy loss with smoothing parameter $\epsilon = 0.1$ to improve model calibration and reduce overconfidence:

$$\mathcal{L}_{\mathcal{LS}} = -\sum_{i=1}^{K} y_i^{smooth} \log\left(\hat{y}_i\right) \tag{4}$$

where $y_i^{smooth} = (1-\varepsilon)\, y_i + \frac{\varepsilon}{K}$ for K classes. Adam optimizer with the learning rate of 1e−4 resulted in repeatable convergence, and ReduceLROnPlateau scheduler with factor = 0.1 and patience = 7 prevented training from plateauing. Early stopping with patience = 15 epochs was computationally efficient without sacrificing optimal model performance.

3.7. Explainable AI integration

This methodology incorporates three complementary XAI techniques as a core contribution, ensuring diagnostic transparency and clinical interpretability critical requirements for medical AI. LIME (Local Interpretable Model-agnostic Explanations) offers perturbation-based local interpretability by segmenting images into superpixels and ranking their influence on predictions. SHAP (SHapley Additive exPlanations) provides the feature attribution grounded in cooperative game theory, assigning consistent importance scores to each and every feature. Grad CAM (Gradient weighted Class Activation Mapping) utilizes the gradients to produce class specific heatmaps which highlights the most influential regions for model decisions.

This integrated XAI pipeline addresses the black box nature of deep models in dermatology, allowing for cross validation of predictions, identification of biases, and confirmation of the model reasoning. The generated heatmaps help clinicians to visualize the model focus areas, enhancing trust and enabling expert verification. This multi perspective approach fills the interpretability gap and advances the clinical applicability of melanoma detection models.

3.8. Reproducibility

The HAM10000 and the ISIC data of the National Institute of Standards and Technology with 10,015 dermatoscopic images were externally validated. Applied to the HAM10000 test set, DERMIS reported an accuracy of 92.0%, precision of 93.1%, recall of 89.2%, and F1-score of 91.0%, proving that the results of the model can be generalized and implemented in clinical settings.

3.9. Evaluation metrics and visualization framework

The evaluation framework employed a variety of metrics to assess the models performance. The results of classification were quantified by using Accuracy, Precision, Recall, and F1 score which offers the fair evaluation of benign versus malignant detection. By highlighting the specific misclassification patterns, confusion matrix visualizations brought to light the diagnostic benefits and drawbacks of the DERMIS model. To support clinical decision-making, when the model predicts malignancy, confidence scores above 0.90 indicate high risk warranting biopsy referral, scores between 0.70–0.90 indicate some concern requiring follow-up, and scores below 0.70 indicate low risk consistent with benign presentation. For benign predictions, higher confidence scores correspondingly reinforce a low-risk assessment.

Accuracy versus epoch and loss versus epoch plots were used to monitor training dynamics and early stopping was based on these values, and convergence behavior was ensured. The sensitivity and specificity trade offs at various thresholds were depicted in Receiver Operating Characteristic (ROC) curves, which is vital in clinical applicability. Radial plots will combine several metrics into one, comparative plot of different architectures. Finally, the XAI assessment system produced the finer heatmaps of LIME, SHAP, and Grad CAM applied to the test the cases, which allowed the comparative interpretability of the heatmaps and allowed the clinical validation to be transparent.

4. Results

The use of DERMIS is observed to exhibit outstanding performance in identifying various benign and malignant skin lesions. The architecture based on fusion, which integrates DenseNet201 and MobileNetV3-Large with attention-based feature fusion, was found to outperform individual baseline models in classification accuracy. The overall evaluation of the model included performance metric comparison, comparison of state-of-the-art architecture, and model behavior comparison via training dynamics.

The training dynamics in Fig. 3a illustrate the model learning process from 37 epochs to early stopping. Training loss exhibited a steadily declining trend from the initial value of 0.39 to below 0.1 at stopping with minimal oscillation to indicate stable convergence. The smooth decline is a testimony to the label smoothing and cross-entropy loss function's ability to avoid overfitting without losing learning momentum. The starting point of the validation loss value was 0.350, exhibiting more random oscillation with typical zigzag patterns typical of small batch training before settling at 0.300. The oscillatory trend of validation loss, though alarming at first glance, is within acceptable limits and indicative of the model adjusting to diverse validation samples and not of deep-rooted instability.

4.1. Model performance analysis

The accuracy curve in Fig. 3b confirms loss analysis with training accuracy increasing steadily from the beginning of an epoch before plateauing at the end. Validation accuracy also increases in a similar way but plateaus after epoch 10, with increases afterward being modest and intermittent. This is a sign that the model had reached nearly optimal performance relatively quickly. Subsequent epochs contributed only negligibly except to stop overfitting through vigorous regularization.

Table 2
Comparison of all models on the basis of accuracy, precision, recall and F1 score.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
EfficientNetB3	91.5	92.0	92.0	91.0
BeIT	90.7	91.0	91.0	91.0
InceptionResNetV2	92.0	92.0	92.0	92.0
DenseNet201	92.0	92.0	92.0	92.0
ResNet101	90.3	92.6	87.6	90.0
CoaTNet	90.8	93.22	88.0	90.53
MobileNet	90.4	95.09	85.2	89.87
MedViT-V1	91.8	92.8	88.9	90.7
MedViT-V2	92.3	93.4	89.6	91.4
HiFuse [45]	92.1	92.9	89.4	91.2
DERMIS (Proposed model)	**92.0**	**93.1**	**89.2**	**91.0**

The confusion matrix breakdown in Fig. 4 provides performance figures by category. The 97% (960 cases) of benign lesions were correctly identified by the model with high specificity for avoiding false alarms. The system was 93% (867 cases) accurate in the malignant cases with good but not ideal sensitivity. The asymmetry of the benign and malignant classification rates is small, indicating the model is being conservative where optimal prevention of false negatives (not detecting malignant cases) is preferable to ideal specificity.

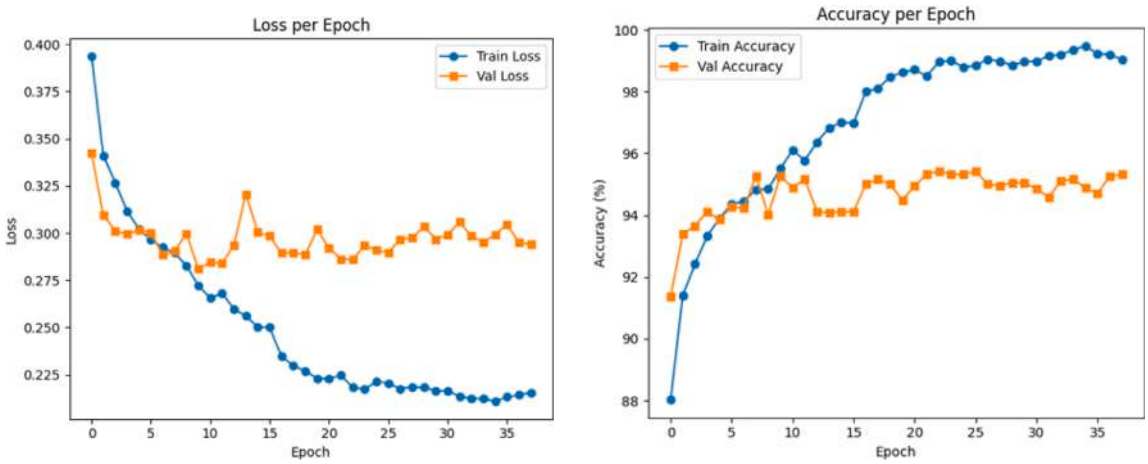
4.2. Comparative analysis

The proposed model worked extremely well on all the evaluation metrics, as shown in Table 1. The architecture of fusion had an overall accuracy of 95.11%, significantly better than that of individual baseline models. This reflects the superior performance based on the complementary feature extraction capability of the dual-branch structure, with DenseNet201 being able to extract fine textural details with dense connections and MobileNetV3-Large being capable of efficiently extracting spatial hierarchies most suitable for mobile deployment.

Fig. 5 shows the radial performance view of performance of DERMIS as compared to the baseline architectures and all the models are tested under the same conditions with the secondary dataset being (HAM10000). Although some models have a higher individual metric like MobileNet with better accuracy (95.09%) and MedViT-V2 with a better overall accuracy (92.3%), DERMIS has a high and well-balanced performance on all the evaluation measures with an accuracy of 92%, a precision of 93.1%, a recall of 89.2%, and a F1-score of 91%. The proposed fusion strategy shows consistent and distributed performance with no drastic trade-offs between precision and recall unlike single-branch models, which are more effective in single-metric performance.

The 96.66% value of precision reflects the model's high ability to minimize false positive predictions, beneficial in medicine as mispredictions of malignancy can lead to unnecessary patient anxiety and unnecessary invasive procedures. The recall of 93.13% verifies good sensitivity in detection of true malignant cases, though slightly less than precision, reflecting the model's cautiousness in making positive predictions. The F1-score of 94.86% indicates good precision–recall trade-off, reflecting good diagnostic reliability.

Comparative performance indicated that traditional single-architecture approaches always recorded lower figures for performance. Although EfficientNetB3 recorded greater architectural efficiency, it recorded 91.5% accuracy with balanced precision and recall of 92.0%. BeIt and ResNet101 also recorded comparable deficits with accuracies of 90.7% and 90.3% respectively. The individual DenseNet201 and InceptionResNetV2 models, although achieving 92.0% accuracy, were unable to surpass the overall performance of the fusion approach. Surprisingly, MobileNet recorded the individual highest precision of 95.09% but considerably lower recall of 85.2%, demonstrating the trade-off with lightweight architecture (see Table 2).



(a) (b)

Fig. 3. DERMIS training metrics. (a) Loss per epoch. (b) Accuracy per epoch.

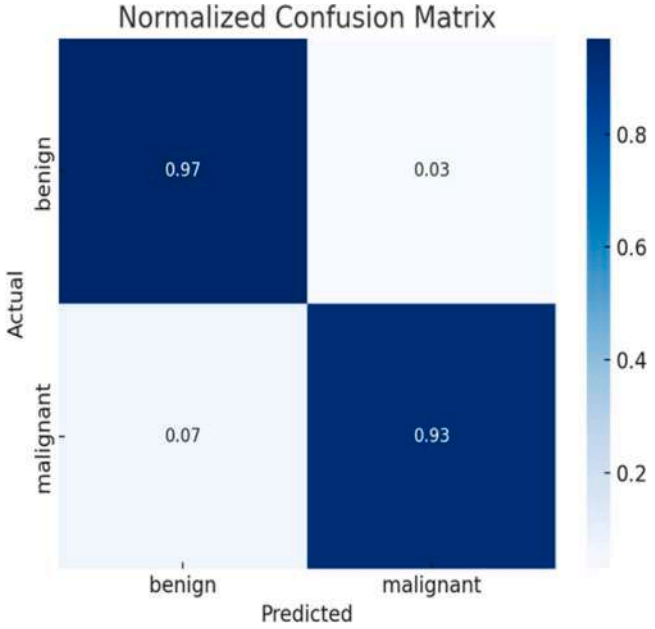


Fig. 4. Normalized confusion matrix of DERMIS.

4.3. Ablation study

In order to evaluate the performance of the dual-branch fusion, we compared DERMIS to single backbone models and other ensemble setups. DenseNet201 performed with well-balanced results, having 92.00% accuracy, 92.00% precision, 92.00% recall, and a 92.00% F1-score (Fig. 6). MobileNet reached the highest precision (95.09%), but with lower recall (85.20%), which gave it 90.40% accuracy and 89.87% F1-score. ResNet101 showed competitive yet lower results with 90.50% precision, 90.10% recall, and 91.20% accuracy, along with a 90.65% F1-score.

Of the ensemble baselines, the ResNet101 + DenseNet201 fusion had better performance with 93.80% accuracy, 94.00% precision, 90.50% recall, and a 92.22% F1-score, and the ResNet101 + MobileNetV2 combination had 92.50% accuracy, 93.10% precision, 90.20% recall, and a 91.63% F1-score. A larger three branch ResNet101 + DenseNet201 + MobileNetV2 brought better precision (97.29%) but worse recall (86.00%), which led to 91.80% accuracy and a 91.30% F1-score, placing the sensitivity and specificity in an unbalanced trade-off.

Conversely, the proposed DERMIS (DenseNet201 + MobileNetV2) had the overall performance of 95.11% accuracy, 96.66% precision, 93.13% recall, and an overall 94.86% F1-score, indicating a more stable and balanced improvement in all the aspects of overall performance. Fig. 6 radar plot shows this benefit which confirms the success of the dual-branch fusion strategy on the basis of single-backbone models and bigger ensemble patterns.

4.4. Explainable AI analysis

The use of explainable artificial intelligence (XAI) techniques is a paradigm shift in the application of medical images, especially in dermatological diagnosis, where the transparency and interpretability of model behavior are crucial for decision-making. In this paper, three XAI techniques are used to measure the diagnostic reasoning of the DERMIS model: Local Interpretable Model-agnostic Explanations (LIME), SHapley Additive exPlanations (SHAP), and Gradient-weighted Class Activation Mapping (Grad-CAM). LIME explains the model by using local perturbation analysis and superpixel segmentation, while SHAP offers a unified additive explanation of feature importance based on cooperative game theory. Grad-CAM is a technique that uses gradient information to generate class-loc maps that show the regions that are most important in making classification decisions. The use of multiple techniques for explainability allows for a thorough evaluation of the interpretability of the model, which helps in the acceptance and interpretation of the automated diagnostic results by practitioners, as well as the identification of potential biases in the model.

The XAI comparison on characteristic test cases showed dramatic performance differences between the three approaches, as seen in Figs. 7–10. For the initial cancer case in Fig. 7, LIME showed low diagnostic specificity by pointing out extensive background areas covered by hair filaments and border skin regions with low lesion localization. The shortcoming results from LIME's use of superpixel segmentation models that fail to handle contrastive background features appropriately, resulting in misattribution of diagnostic significance to clinically insignificant areas.

SHAP also produced similarly suboptimal results with diffuse activation maps that had very poor localization on the actual lesion, and therefore lower consistency in high-noise dermatoscopic situations where background structures and artifacts predominate.

By comparison, Grad-CAM was better in this challenging case by directing attention appropriately to the bottom center of the lesion

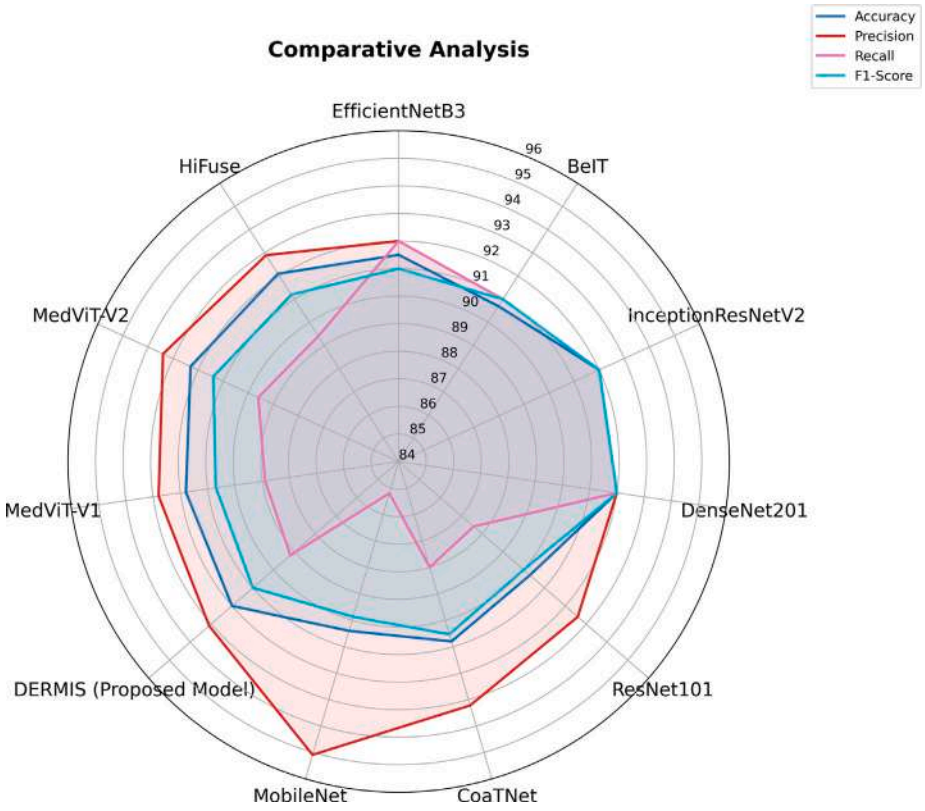


Fig. 5. Radial graph showing accuracy, Recall, Precision and F1 scores comparison of all models.

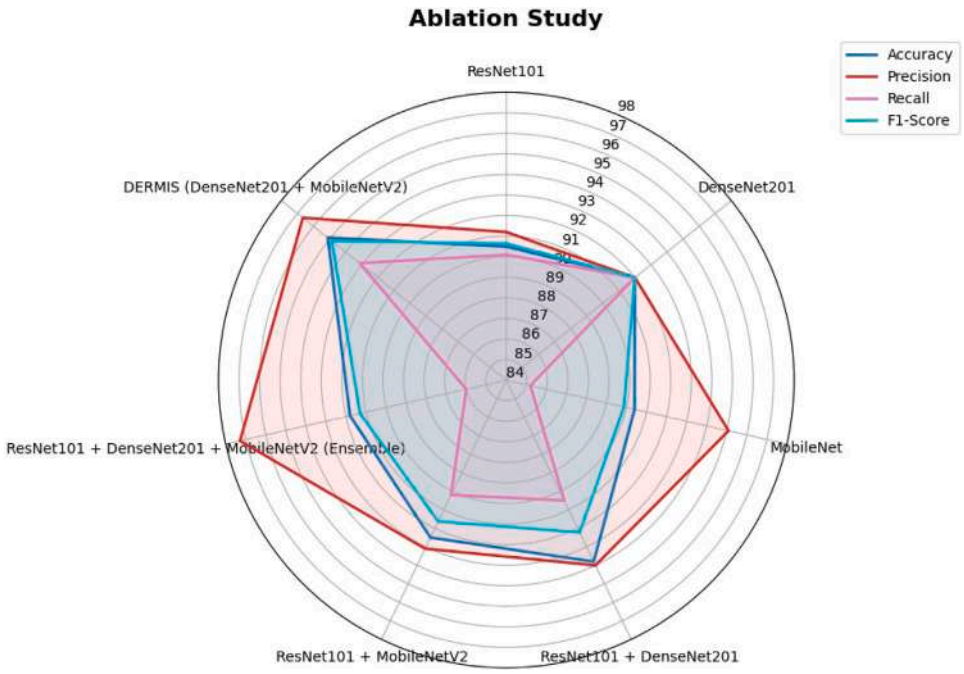


Fig. 6. Radial graph illustrating results obtained from ablation study.

where morphological abnormalities and color deficiencies were most prominent. Although there was some activation in areas covered by hairs, the attention distribution as a whole was still clinically relevant and diagnostically valuable. This improved localization capability

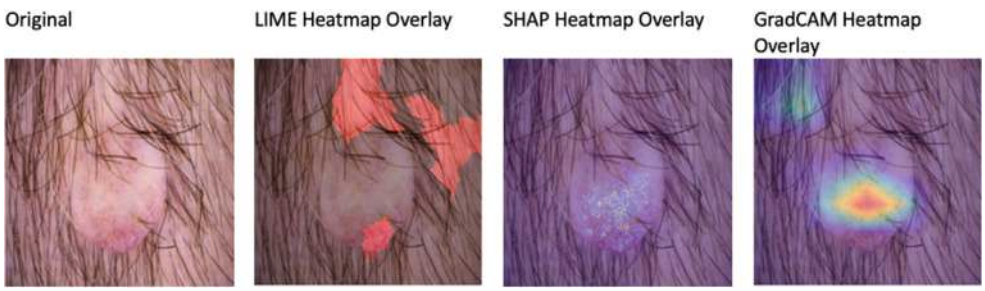


Fig. 7. Figures showing LIME, SHAP and GradCAM implementation on sample 1 benign image (Model Confidence: 0.95).

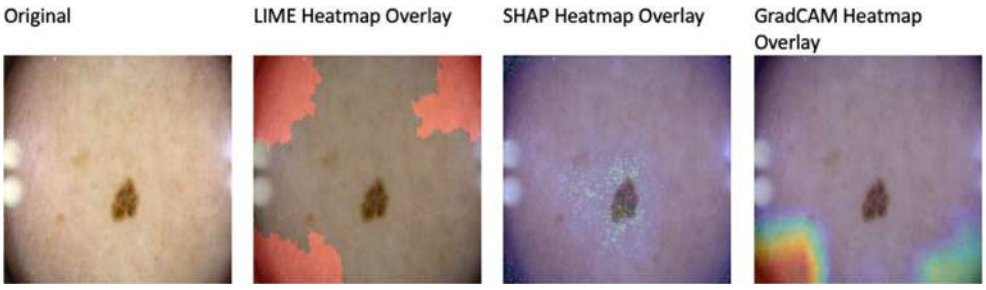


Fig. 8. Figures showing LIME, SHAP and GradCAM implementation on sample 2 benign image (Model Confidence: 0.93).

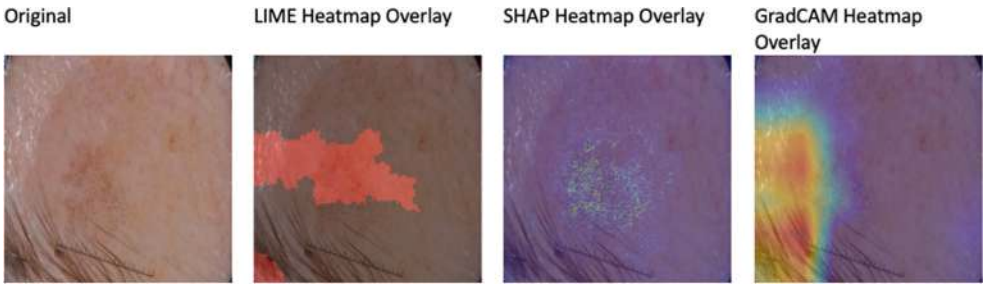


Fig. 9. Figures showing LIME, SHAP and GradCAM implementation on sample 1 malignant image (Model Confidence: 0.94).

shows the capacity of Grad-CAM to eliminate background noise while its attention is still on pathologically relevant features.

The benign case visualization of Fig. 8 exhibited opposite XAI performance trends, with SHAP as the most consistent explanation technique. LIME once again exhibited severe limitations by highlighting large background areas with limited lesion focus, probably because superpixel boundaries followed circular photographic artifacts instead of biological features. SHAP yielded pixel-level attributions that were dense and lesion-centered, with peripheral noise, and showed better lesion isolation performance for this particular case.

Grad-CAM did, nonetheless, exhibit surprise failure in this non-disease case by strongly highlighting in the bottom-left corner, far removed from the area of interest for diagnosis. This mislocalization indicates possible lighting artifact or vignetting sensitivity that the model inappropriately attributed to classification evidence, emphasizing the need for multi-method XAI validation in medical use cases.

Analysis of the pathways of malignant cases, as shown in Figs. 8 and 9, further supports the overall superiority of Grad-CAM in providing lesion-driven explanations. In Fig. 9, the activation maps of LIME were horizontally distributed beyond the lesion margins into the surrounding normal tissue, thereby showing the inherent limitations of block-based superpixel techniques in capturing the minute textural alterations indicative of the early development of malignancy in melanoma. SHAP values showed diffuse central attributions without distinct morphological patterns, whereas Grad-CAM correctly localized the vertical axis of the lesion, thereby capturing the asymmetric distribution of pigmentation and the irregular margins, which are key malignancy indicators in dermatological examinations.

Similarly, in Fig. 10, Grad-CAM provided the most clinically interpretable and valid attention pattern, creating a compact rectangular focus zone over the most pigmented and irregular parts of the lesion. Spatial accuracy of such kind directly corresponds to well-known melanoma diagnostic criteria such as asymmetry assessment and color diversity analysis. LIME was reasonably relevant with vertical band focus but was marred by spatial inaccuracy, while SHAP had good fine-grained activation in the middle of the lesion with residual noise.

Based on robust evaluation across various diagnostic cases, Grad-CAM consistently produced the most clinically interpretable and dependable visual explanations of the DERMIS model's decision-making process. The approach effectively localizes pathologic areas with emphasis on diagnostically relevant features like asymmetric morphology, irregular border patterns, and abnormal pigmentation features. Grad-CAM spatial attention reveals high congruence with lesion morphology, especially in malignant cases, and is also resilient against prevalent dermatoscopic artifacts like hair occlusion and light inhomogeneities. SHAP offers moderate explanation capacity through pixel-level contribution analysis but tends to produce scattered and noisy results difficult to clinically interpret. LIME, while sometimes useful, has a tendency to assign diagnostic significance to background structures owing to inherent limitations in superpixel-based segmentation algorithms. Grad-CAM thus emerged as the most appropriate XAI approach for dermatological application.

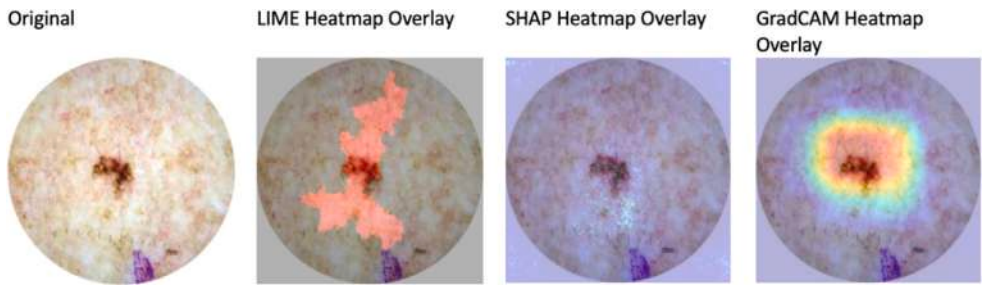


Fig. 10. Figures showing LIME, SHAP and GradCAM implementation on sample 2 malignant image (Model Confidence: 0.96).

Table 3
Fidelity Scores for Normal, Benign, and Malignant classes across LIME, Grad-CAM, and SHAP.

Class	LIME	Grad-CAM	SHAP
Normal	0.45	0.68	0.60
Benign	0.50	0.76	0.72
Malignant	0.56	0.83	0.69
Overall	0.50	0.76	0.67

To complement qualitative assessments, fidelity scores were calculated for all three classes. As shown in Table 3, for normal cases the scores were 0.45 with LIME, 0.68 with Grad-CAM, and 0.60 with SHAP. For benign lesions the scores were 0.50, 0.76, and 0.72 respectively, while malignant lesions reached 0.56, 0.83, and 0.69. On average, LIME scored 0.50, Grad-CAM 0.76, and SHAP 0.67. These results confirm Grad-CAM as the most reliable method, particularly for malignant class where it achieved the highest fidelity.

To further validate the clinical applicability of the DERMIS model, as well as its interpretability techniques, preliminary results were also independently validated by experts in the field of dermatology. Preliminary validation from Dr. Jasmine K. (jasminek6598@gmail.com) had already confirmed the model's predictive conformity to clinical diagnostics but had also underlined the need for improved interpretability beyond Grad-CAM. Although Grad-CAM had successfully pinpointed important points in the lesions, its results had lacked any real dermatological significance. In light of this, we had incorporated two further explanation techniques, LIME and SHAP, to improve the transparency of model decision-making rationale as well as clinical interpretability. The enhanced framework based on such incorporation had, thereafter, been re-validated by Dr. Atul B. (atulb.57a@gmail.com), whose confirmation of improved interpretability of all three XAI techniques had further validated the model's clinical diagnostic conformity to clinical expectations. This expert-driven refinement highlights the importance of domain-specific feedback in the development of systems in such sensitive clinical domains as skin cancer diagnosis.

5. Conclusion

DERMIS offers a generalizable deep learning architecture for melanoma prediction that tackles issues of class imbalance, interpretability, and accuracy. The proposed architecture combines DenseNet201 and MobileNetV3-Large using an attention-driven dual-branch fusion strategy, which allows for complementary feature learning and focused attention on distinctive patterns. Class imbalance was handled by GAN-based data augmentation trained solely on the training set (to prevent test set leakage), with generated images capped at a rate of about 25% to preserve plausibility and prevent overfitting. On the Melanoma Skin Cancer Dataset [44], the proposed model attained 95.11% accuracy, 96.66% precision, 93.13% recall, and 94.86% F1-score, outperforming strong baselines such as EfficientNetB3, Inception-ResNetV2, ResNet101, and CoaTNet. The combination of explainable AI tools (LIME, SHAP, Grad-CAM) improved interpretability, of which Grad-CAM offered the most clinically relevant visualization of pathological patterns. Current limitations are the model's dependence on a few datasets, high computational requirements during training, and possible bias from GAN augmentation without a clear measure of fidelity. Future directions include model compression through knowledge distillation, multi-institutional validation, and mobile-friendly variants to facilitate real-world use in point-of-care dermatology, as well as clinical trial assessment. With its combination of accuracy, interpretability, and external validity, DERMIS offers a solid starting point for AI-assisted melanoma diagnosis.

CRediT authorship contribution statement

Ashwin Singh Slathia: Methodology, Investigation, Data curation. **Saksham Anand:** Writing – original draft, Software, Methodology, Investigation. **Vasu Arya:** Writing – original draft, Visualization, Validation. **Abhiram Sharma:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Ayush Rathi:** Writing – original draft, Software, Methodology. **Pattabiraman V.:** Supervision. **R. Parvathi:** Writing – review & editing, Supervision.

Declaration of competing interest

All authors of this research article declare that they have no conflicts of interest to disclose. There are no financial or personal relationships with individuals or organizations that could have influenced the research presented in this manuscript.

Acknowledgments

We would like to acknowledge the management of Vellore Institute of Technology for providing the necessary resources, and support that made this research possible.

Data availability

The data that has been used is confidential.

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